

Pakhi Banchalia

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RESEARCH INTERESTS

AI Safety, Alignment, Reinforcement Learning, Robustness, Multi-objective Optimization, Biologically Inspired Learning

RESEARCH EXPERIENCE

- **Aalto University** June 2026 - Present
Summer Research Intern Espoo, Finland
 - Exploring social intrinsic motivation in multi-agent reinforcement learning with [Prof. Guckelsberger](#)
- **Lossfunk** July 2025 - Sept 2025
Research Intern Bangalore, India
 - Built and evaluated homeostatic RL agents in Crafter to study behavior under competing internal drives
 - Analyzed how reward structure variations affect exploration, behavioral stability, and risks of unintended or misaligned policies
 - Compared homeostatic agents to standard RL baselines to evaluate behavioral diversity, robustness, and failure modes under shifting reward structures
- **Computational Machinery of Cognition (CMC) Lab, TU Dresden** Nov 2024 - June 2025
Research Intern (Independent Collaboration) Dresden, Germany
 - **Homeostatic Reinforcement Learning:** Designed RL agents with internal physiological drives to study generalization and retention; analyzed how internal regulation affects long-term stability and generalization under changing task distributions.
 - **Dopamine (Reward-Modulated Learning):** Implemented a biologically inspired gradient-free learning algorithm using reward-modulated plasticity; investigated credit assignment without backpropagation via reward-modulated weight perturbations.
- **Google Summer of Code, INCF** May 2024 - Sept 2024
Contributor Remote
 - Simulated *C. elegans* neural development using Neural Cellular Automata and evolutionary optimization
 - Modeled evolving neural connectivity using Graph Neural Networks
 - Applied Topological Data Analysis (Kepler Mapper) to analyze developmental and connectivity trajectories
 - Delivered open-source, reproducible [code](#) for developmental neural modeling

PUBLICATIONS

- Pillai, H.S., **Banchalia, P.**, Jamboji, V., Krishnan, S.R. (2026). *CAN: Continuously Adapting Network*. In *Data Science and Applications, ICDSA 2025*. Lecture Notes in Networks and Systems, vol 1723. Springer, Cham. [\[DOI\]](#)

RESEARCH PROJECTS

- **Homeostatic Crafter**
 - Designed agents governed by multiple internal drives (food, energy, safety, etc.) influencing decision-making
 - Studied how internal regulation affects exploration and behavioral diversity
 - Investigated modular critic architectures for stabilizing multi-objective RL
- **Dopamine: Reward-Modulated Gradient-Free Learning**
 - Contributed to biologically inspired learning rules using reward-modulated weight perturbation
 - Explored alternative credit assignment mechanisms without gradient backpropagation
 - Demonstrated competitive performance in early benchmarks
- **SEANN: Self-Expanding and Adaptive Neural Networks**
 - Extended the CAN continual learning framework with dynamically expanding network capacity
 - Addressed static-capacity limitations by enabling task-relevant subnetworks during continual learning
 - Explored structural plasticity as an alternative to fixed model scaling and mitigation of catastrophic forgetting

TALKS AND POSTERS

- **Brain-like features in homeostatic deep reinforcement learning agents @ Bernstein Conference 2025**
 - Poster presentation on homeostatic RL agents integrating internal and sensory inputs, showing structured behavioral and neural patterns
- **Emergence of Brain-like Features in Homeostatic RL Agents @ BrainBody 2025**
 - Oral presentation; demonstrated coupling between internal states, neural dynamics, and adaptive exploration

TECHNICAL SKILLS

- **Languages:** Python, C++, MATLAB, SQL
- **ML / Research:** PyTorch, TorchVision, Gymnasium (OpenAI Gym), JAX (familiar)
- **Scientific Computing:** NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper (TDA)
- **Research Areas:** Reinforcement Learning, Biologically-Inspired Learning, Neural Cellular Automata, Continual Learning, GNNs
- **Developer Tools:** Git, Linux/Unix, Docker, LaTeX, Weights Biases (WB)

EDUCATION

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|--|---------------------------|
| • Technische Universität Dresden | <i>Oct 2025 - Present</i> |
| <i>Master of Science in Computational Modeling and Simulation (Applied AI track)</i> | Dresden, Germany |
| • Amrita Vishwa Vidyapeetham Kerala, India | <i>2021-2025</i> |
| <i>Bachelor of Technology in Computer Science and Artificial Intelligence</i> | Amritapuri, India |